MultiChallenge: A Realistic Multi-Turn Conversation Evaluation Benchmark Challenging to Frontier LLMs

Ved Sirdeshmukh,[*] Kaustubh Deshpande,[*] Johannes Mols,[*] Lifeng Jin,
Ed-Yeremai Cardona, Dean Lee, Jeremy Kritz,
Willow Primack, Summer Yue, Chen Xing
Scale AI

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Abstract

We present MultiChallenge, a pioneering benchmark evaluating large language models (LLMs) on conducting multi-turn conversations with human users, a crucial yet underexamined capability for their applications. MultiChallenge identifies four categories of challenges in multi-turn conversations that are not only common and realistic among current human-LLM interactions, but are also challenging to all current frontier LLMs. All 4 challenges require accurate instruction-following, context allocation, and in-context reasoning at the same time. We also develop LLM as judge with instance-level rubrics to facilitate an automatic evaluation method with fair agreement with experienced human raters. Despite achieving near perfect scores on existing multi-turn evaluation benchmarks, all frontier models have less than 50% accuracy on MultiChallenge, with the top-performing Claude 3.5 Sonnet (June 2024) achieving just a 41.4% average accuracy.

1 Introduction

Evaluation benchmarks have played a critical role in the evolution of large language models (Ziyu et al., 2023; Porcu and Havlínová, 2024), with benchmarks being continuously released to test large language models (LLMs) on various tasks (Zhang et al., 2024; Jain et al., 2024; Glazer et al., 2024). One such task, conducting multi-turn conversations with human users, is a common yet challenging task for LLMs. It is challenging because conducting multi-turn conversations requires not only accurate instruction following, but also careful attention allocation to conversation context, and more importantly, near or surpassing human in-context reasoning capability at the same time.

Despite the growing demand and complexity of multi-turn LLM conversations with humans, there are limited comprehensive evaluation frameworks (Kwan et al., 2024; Wang et al., 2023) designed for this. Some of them, such as the widely adopted MT-Bench (Zheng et al., 2023), are saturated by frontier LLMs with near-perfect results (Achiam et al., 2023; Yang et al., 2024). Others (He et al., 2024) focus more on multi-turn explicit instruction following, missing the opportunity to assess the actual set of mixed model capabilities required in conducting natural multi-turn conversations with human users.

To bridge this gap, we build MultiChallenge, a realistic and challenging multi-turn conversation benchmark that evaluates models' capability to properly respond to human users given multi-turn conversation histories. Each test example in MultiChallenge is a maximum 10-turn conversation history between two parties, ending with a final user turn containing a requirement/question. LLMs are required to respond to the final user turn properly given the multi-turn conversation history. We identify four categories of challenges in multi-turn conversations that are not only common and realistic among current human-LLM interactions, but also difficult for current frontier LLMs. They include instruction retention, inference memory of user information, reliable versioned editing, and self-coherence.

Instruction retention evaluates whether LLMs are able to follow instructions specified in the first user turn throughout the entire multi-turn conversation. Inference memory of user information evaluates LLMs on recalling and connecting relevant details scattered in previous user turns when they are implicitly required to respond to the final user turn. Reliable versioned editing evaluates whether LLMs can properly help humans revise existing materials through back-and-forth iterations with human users. Finally, self-coherence evaluates whether LLMs can be reasonably coherent with model responses in the conversation history and

[*] Indicates Equal Contribution

avoid sycophancy (unconditionally agreeing to human users). We elaborate on the 4 challenges in Section 3 and show examples of them in Figures 1 and 2. All 4 challenges require not only instruction following, but accurate context allocation and in-context reasoning at the same time.

We adopt a hybrid approach in which human raters and LLMs work together, to produce diverse, realistic, and challenging multi-turn test examples. We also develop an LLM-as-a-judge system with instance-level rubrics to facilitate an automatic evaluation method with good agreement with experienced human raters. Evaluation results with both human and automatic evaluation show that all frontier models have less than 50% accuracy on `MultiChallenge`, despite reaching near perfect scores on existing multi-turn evaluation benchmarks (Achiam et al., 2023; Yang et al., 2024), indicating the difficulty and usefulness of `MultiChallenge` in multi-turn conversation evaluation. Among them, Claude 3.5 Sonnet (June 2024) achieves the best general performance, reaching a 41.4% average accuracy score on `MultiChallenge`, significantly outperforming other LLMs. Our experiments also show high correlation between human raters and our automatic judgments with instance-level rubrics, demonstrating the potential for quick and accurate multi-turn evaluations. We open source the data and code of this benchmark at `https://github.com/ekwinox117/multi-challenge`.

2 Related Work

Evaluation benchmarks on multi-turn conversations. Compared to the fast emerging evaluation benchmarks assessing other frontier capabilities of LLMs, such as benchmarks for AI as agent (Liu et al., 2023; Yao et al., 2024) and multimodality (Liu et al., 2025; Li et al., 2024), evaluation benchmarks on multi-turn conversations have not received enough attention. Existing benchmarks, such as MT-Bench (Zheng et al., 2023) and MT-Eval (Kwan et al., 2024), although widely adopted before, are not discriminative for frontier LLMs now as model capabilities keep improving. Frontier models such as GPT-4, Claude 3 and LLama3, all achieve near perfect scores on these benchmarks. More recently, multi-IF (He et al., 2024) has been built to extend the IFEval benchmark (Zhou et al., 2023) to multi-turn and multilingual scenarios. It focuses more on the auto-verifiable format and instruction following capability in multi-turn settings, such as keeping all upper cases, instead of the mixed set of model capabilities required in realistic multi-turn conversations, such as attention allocation of context and in-context reasoning.

Multi-agent synthetic data generation. Multi-agent frameworks has emerged as a popular method for creating diverse, high-quality synthetic datasets. AgentInstruct (Mitra et al., 2024) demonstrated this approach, using multiple LLM-powered agents in three agentic flows to generate synthetic text data with minimal human intervention. Other similar frameworks adopt multi-agent system to generate data for different use cases and knowledge domains, such as Splunk's MAG-V (Sengupta et al., 2024) for customer data and MATRIX (Tang et al., 2024) for social dynamics. In this work, we also adopt a multi-agent method to first generate synthetic multi-turn conversations for all 4 challenges in `MultiChallenge` and then use a hybrid approach to edit and review such synthetic data to produce the final benchmark.

3 `MultiChallenge`

In `MultiChallenge`, we focus on four categories of common and realistic challenges that arise in multi-turn human-LLM interactions that are also difficult for current frontier LLMs. For each category of challenges, we release various realistic test examples curated according to definition of the corresponding challenge. We show the statistics of `MultiChallenge` in Table 1. In this section, we first introduce the detailed definitions of the four challenge categories in `MultiChallenge`. Then we describe the automatic evaluation method that we provide to facilitate efficient and trustworthy adoption of `MultiChallenge`.

3.1 Challenge Categories

3.1.1 Instruction Retention.

When human users interact with LLMs, it is common for human users to specify general instructions at the beginning of the conversation and expect LLMs to follow them throughout the conversation. While in practice, we find that although frontier LLMs are good at following instructions in single-turn conversations, their capability to retain instructions in multi-turn conversations is still limited, which is what we evaluate under Instruction Retention. The first panel in Figure 1 shows

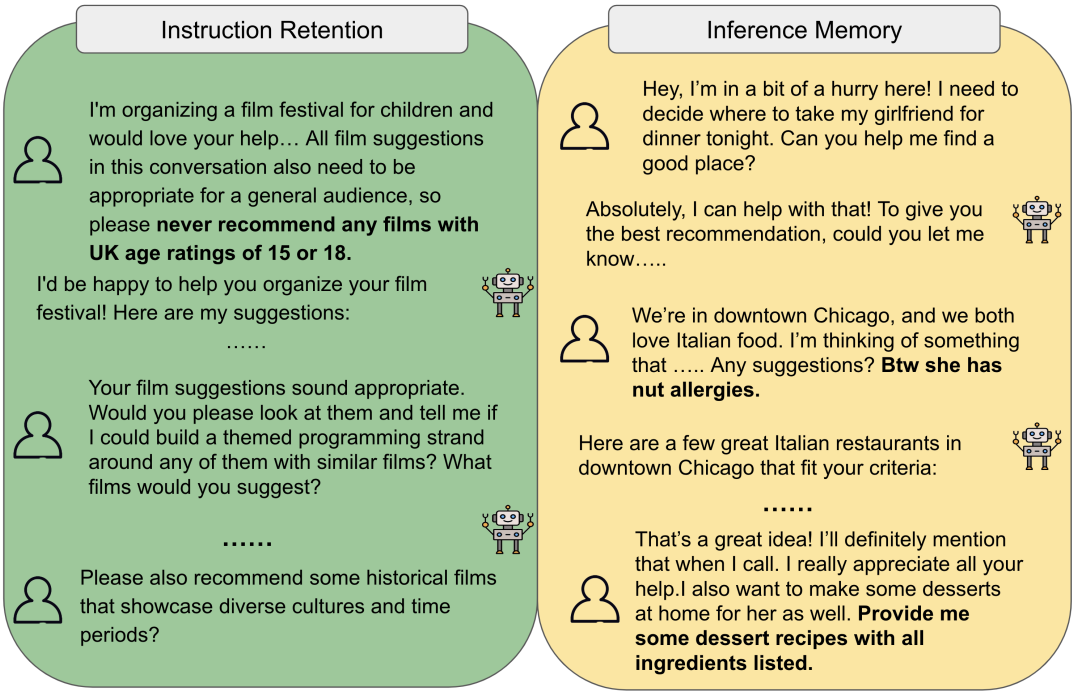

Figure 1: Test examples under Instruction Retention and Inference Memory. The highlighted text under Instruction Retention is the first-turn constraint that should be followed by LLMs for the whole conversation. The highlighted text under Inference Memory is the user information that should be recalled and connected when responding to the final user turn. See Sections 3.1.1 and 3.1.2 for details.

an example. In this example, the constraint about films with specific UK age ratings is given in the first turn. The model is evaluated on its response to the final user utterance still being compliant to the constraint.

In every instruction retention test example, the human user explicitly specify in the first turn that the instructions should be followed throughout the entire conversation. Moreover, human users don't provide followup instructions that contradict the first-turn instructions. The first-turn user instructions also range from simple format constraints such as sentence numbers and bullet point formats, to more semantic instructions such as movie ratings in the example above.

3.1.2 Inference memory (of user information).

Test conversations under this category mainly assess an LLM's capability to recall and connect relevant details scattered in previous user turns when they are implicitly required to respond to the final user turn. The second panel of Figure 1 shows an example of this category. Some user information that appeared early in the conversation, *nut allergies*, is implicitly required to successfully complete the request in the last turn, *dessert recipes*. In such examples, the LLM is evaluated on whether it is able to reallocate its attention to the relevant parts of the conversation history and provide *dessert recipes* that avoid triggering *nut allergies*.

When creating test examples under inference memory category, we pay close attention to make sure that the final user turns don't directly ask for user information mentioned previously, such as "does my girlfriend have nut allergies". Instead, the final user turns implicitly require such user information. Through this way, we test LLMs' capability of reasoning on the relevance of context information.

3.1.3 Reliable versioned editing.

Another common and natural use case of LLMs is to help humans revise existing materials, such as travel plans, emails, code, etc. Such editing usually requires multiple turns of back and forth between the human user and LLMs. These multi-turn conversations usually include multiple versions of LLM-edited materials following different user instructions. It is also common for users to change opinions frequently and give up current edits and move back to editing earlier versions. The first panel in Figure 2 shows a test example, where the user adjusted the schedule multiple times and referred to different versions of it. This collabora-